

Abdul Wahid

Lab Report

Threat Intelligence & Reporting — IBM X-Force Exchange, KeepNote & Word

Threat Intelligence Using IBM X-Force Exchange

IBM X-Force Exchange is a cloud-based threat intelligence sharing platform that enables analysts to collect, analyze, and disseminate threat intelligence. This lab demonstrates data collection, threat analysis, and threat intelligence sharing using IBM X-Force Exchange.

Task 1: Data Collection using IBM X-Force Exchange

Step 1: Registering with IBM X-Force Exchange

Navigated to exchange.xforce.ibmcloud.com and initiated registration for IBM X-Force Exchange by creating an IBMid account.

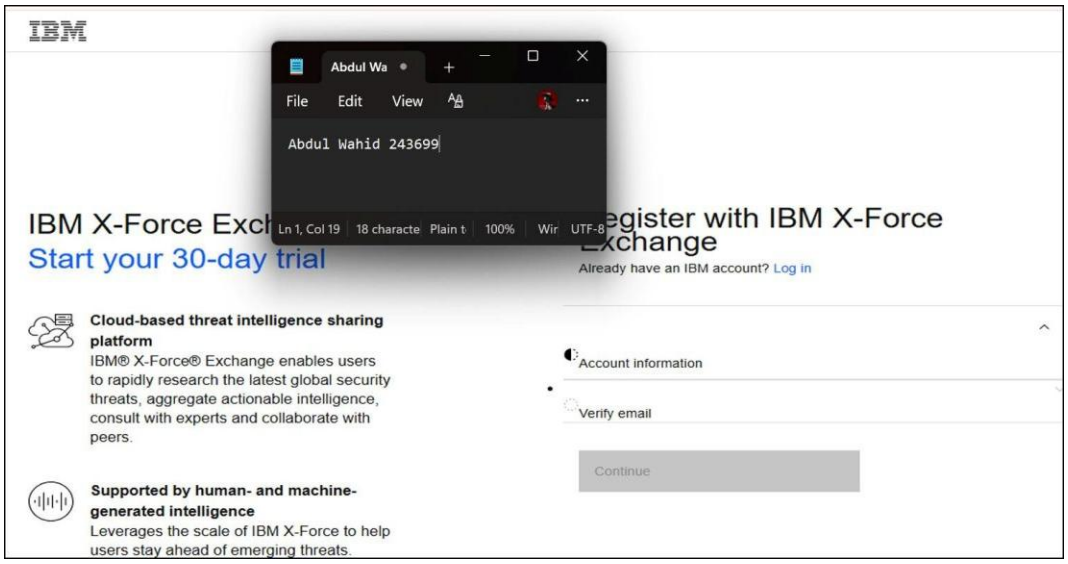


Fig 1.1 – IBM X-Force Exchange Registration Page

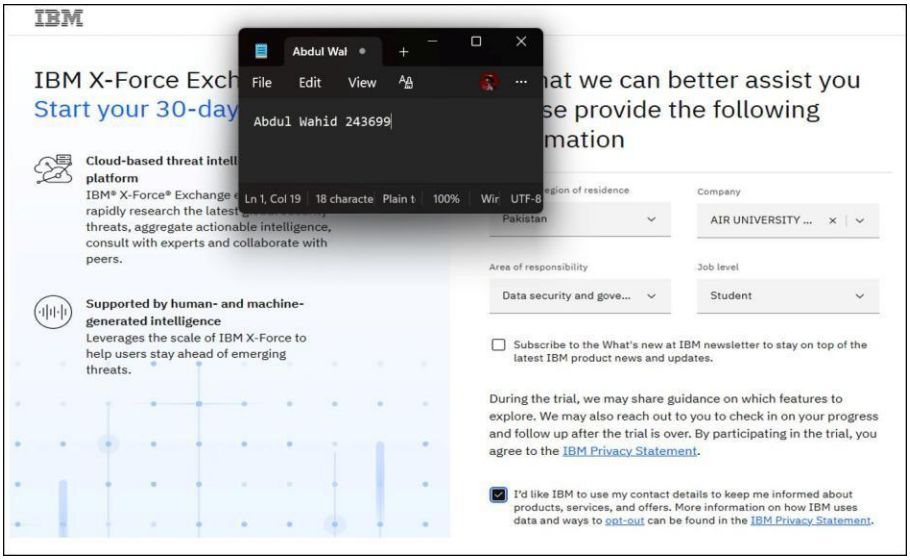


Fig 1.2 – Filling in Registration Details (Country: Pakistan, Company: AIR UNIVERSITY, Role: Student)

Step 2: Logging into IBM X-Force Exchange Dashboard

After email verification, successfully logged in to the IBM X-Force Exchange Dashboard. The dashboard displays trending indicators, recent advisories, threat activity, vulnerabilities, and botnet distribution.

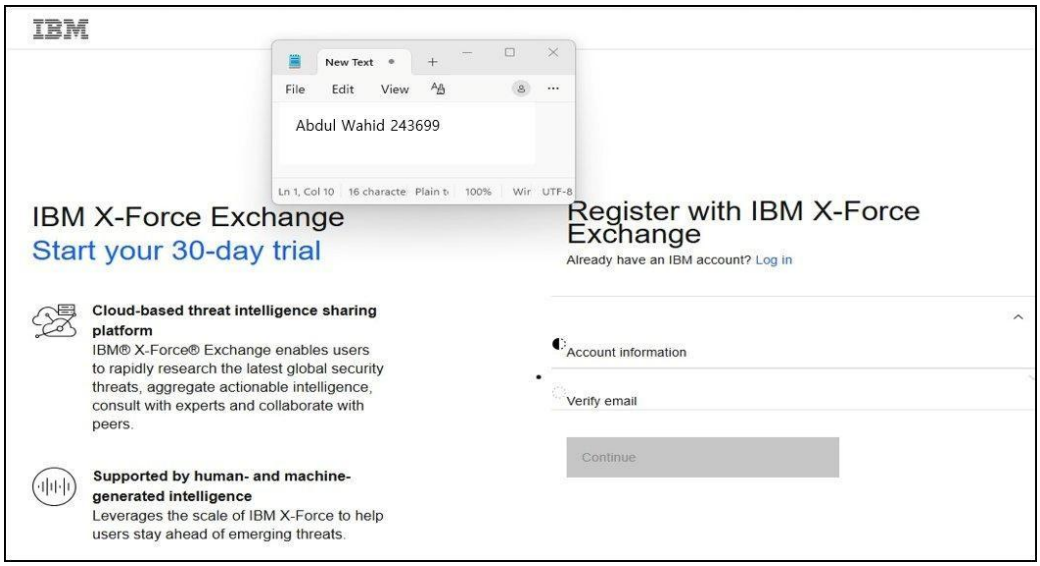


Fig 1.3 – IBM X-Force Exchange Dashboard (Logged In)

Step 3: Exploring IBM X-Force Advisories – New Tactics Seen In Smoke Loader

Clicked "View more" under Recent IBM X-Force Advisories and opened the collection "New Tactics Seen In Smoke Loader" to review Indicators of Compromise (IoCs). The collection includes SHA-256 hashes and IPv4 addresses associated with this threat.

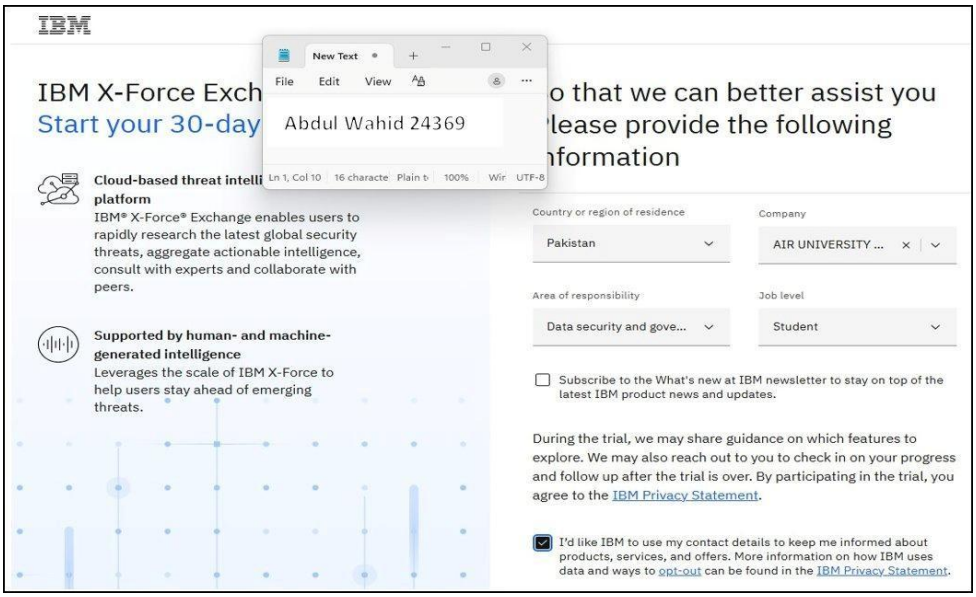


Fig 1.4 – New Tactics Seen In Smoke Loader – Indicators of Compromise (SHA-256 & IPv4)

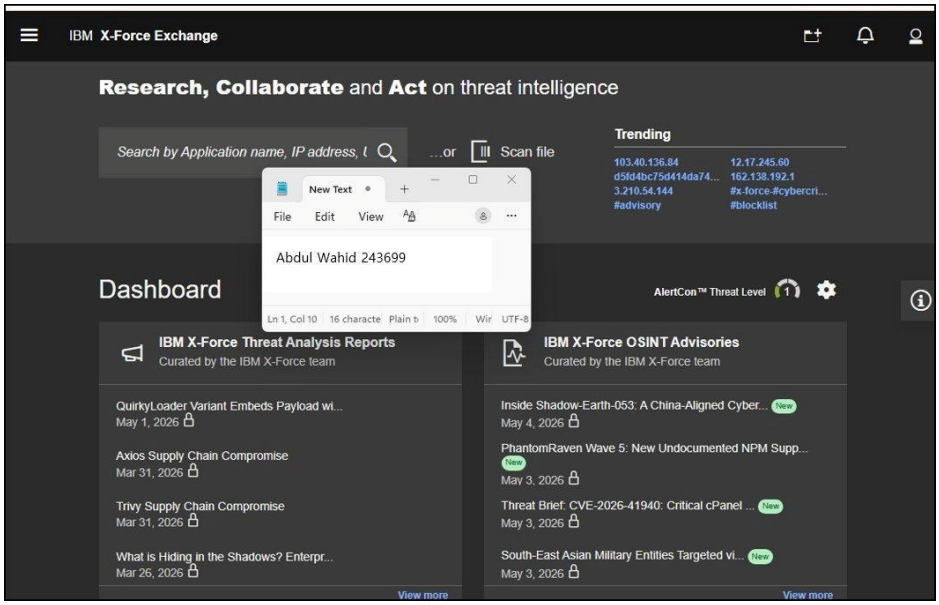


Fig 1.5 – Collection Details Panel Showing Reports (28 Reports)

Step 4: Viewing All Reports for the Collection

Clicked "View all reports" to see the complete list of 28 reports associated with this threat collection, including risk level, type, IOC addresses, and dates added.

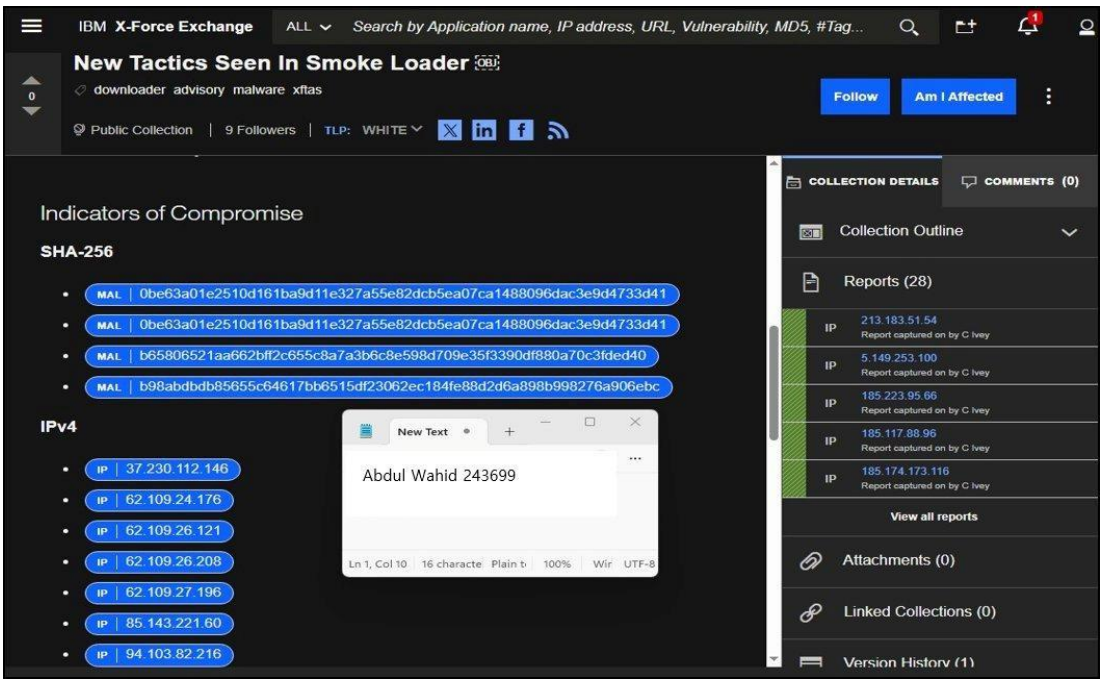


Fig 1.6 – Reports Page: All 28 Reports with Risk Level, Type (IP), IOC, Added By, Date Added

Step 5: Current Malicious Activity Map

Navigated to the Current Malicious Activity map (exchange.xforce.ibmcloud.com/activity/map) to visualize real-time global threat activity including malicious IP addresses by geographic location.

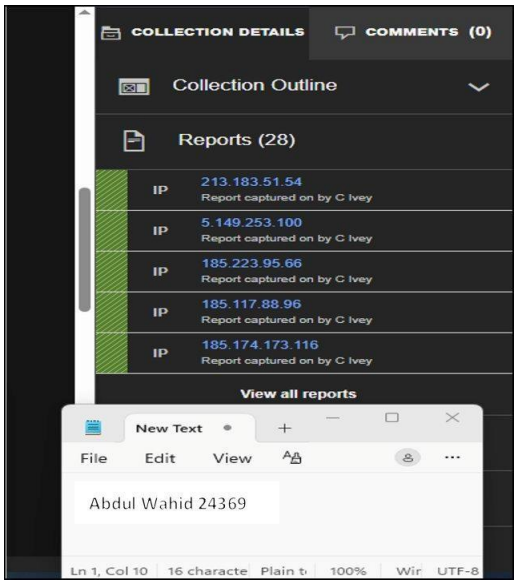


Fig 1.7 – Current Malicious Activity World Map

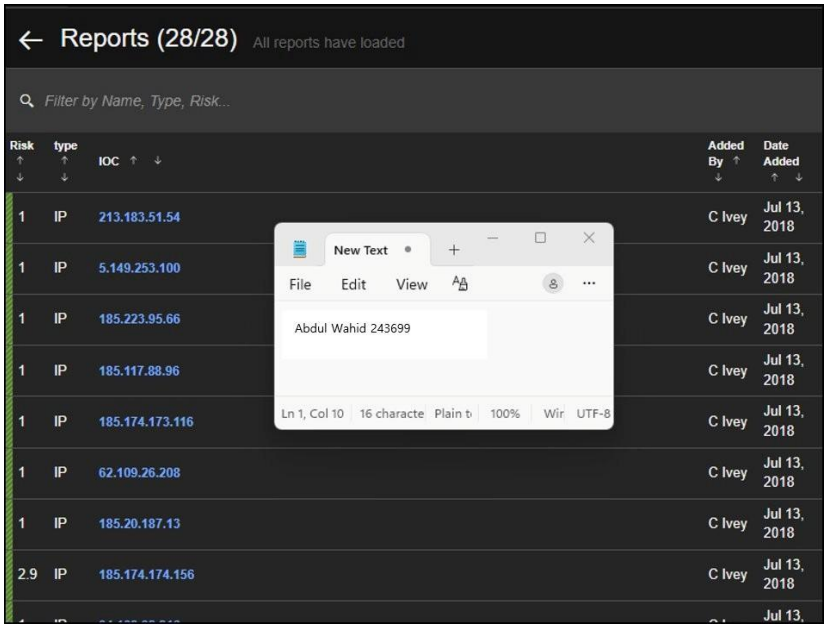


Fig 1.8 – Current Malicious Activity Map (Full View)

Step 6: Vulnerabilities – Activity Timeline

Scrolled down in the Dashboard and clicked "View more" under Vulnerabilities to see the Activity Timeline listing recently identified vulnerabilities with their CVE details.

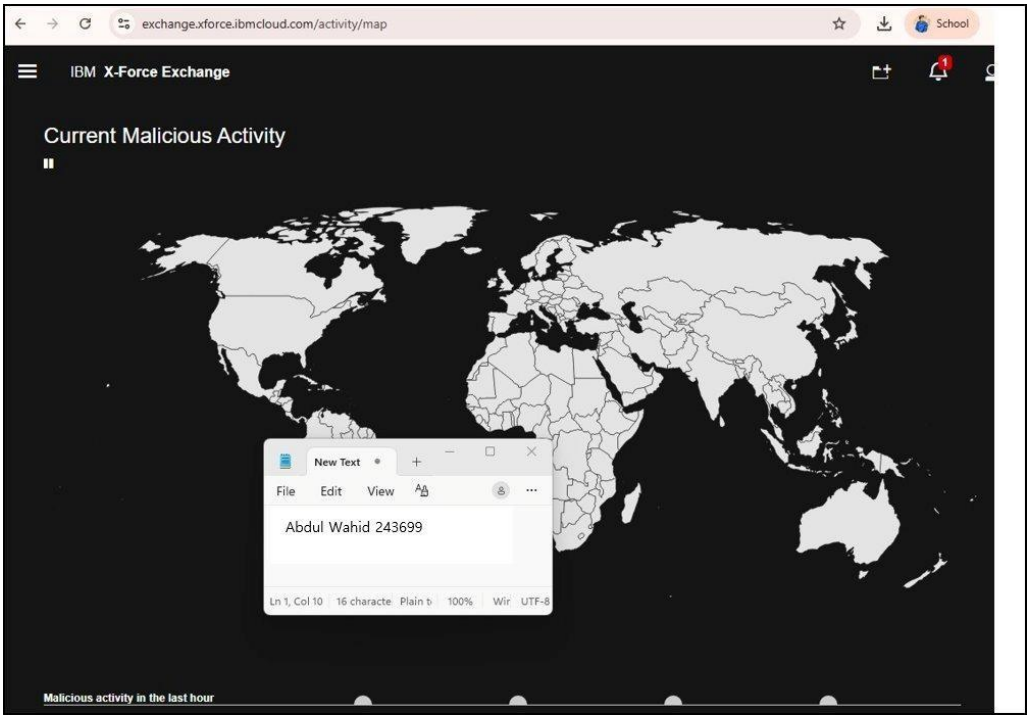


Fig 1.9 – Vulnerabilities Section in Dashboard

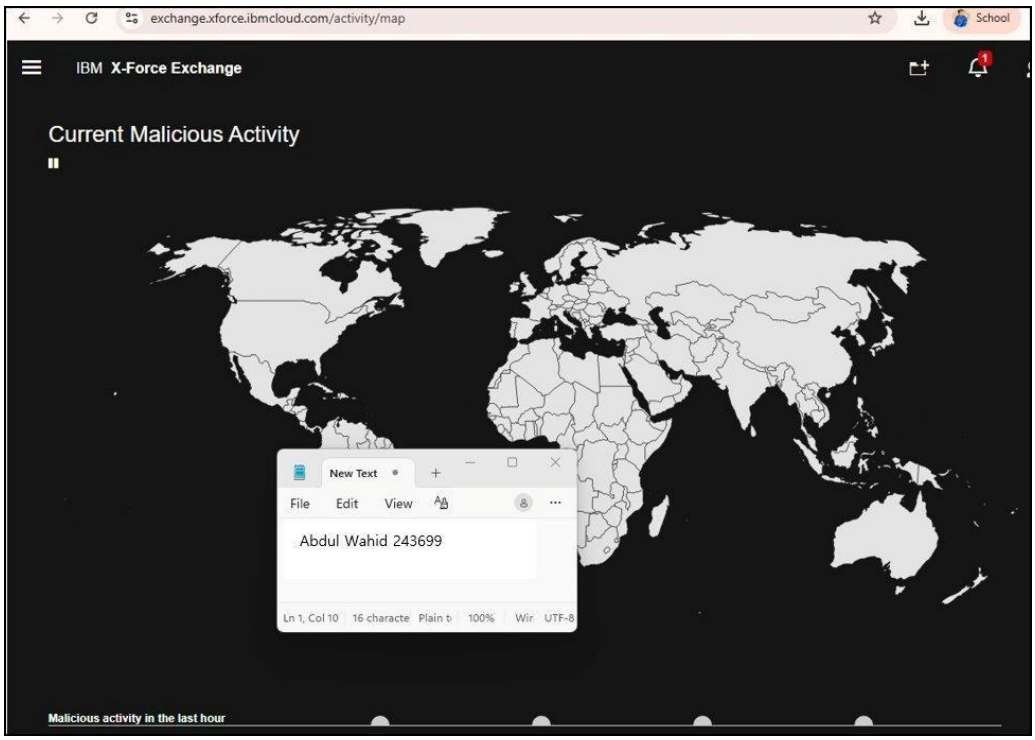


Fig 1.10 – Activity Timeline Showing New Vulnerabilities with CVE Information

Step 7: Viewing a Specific Vulnerability Report

Clicked on the vulnerability "nginx-ui: Unauthenticated First-Run Installer Allows Remote Initial Admin Claim (CVE-2026-42221)" to view its CVSS 3.1 Base Score (8.1 – High), attack vector details, and remediation information.

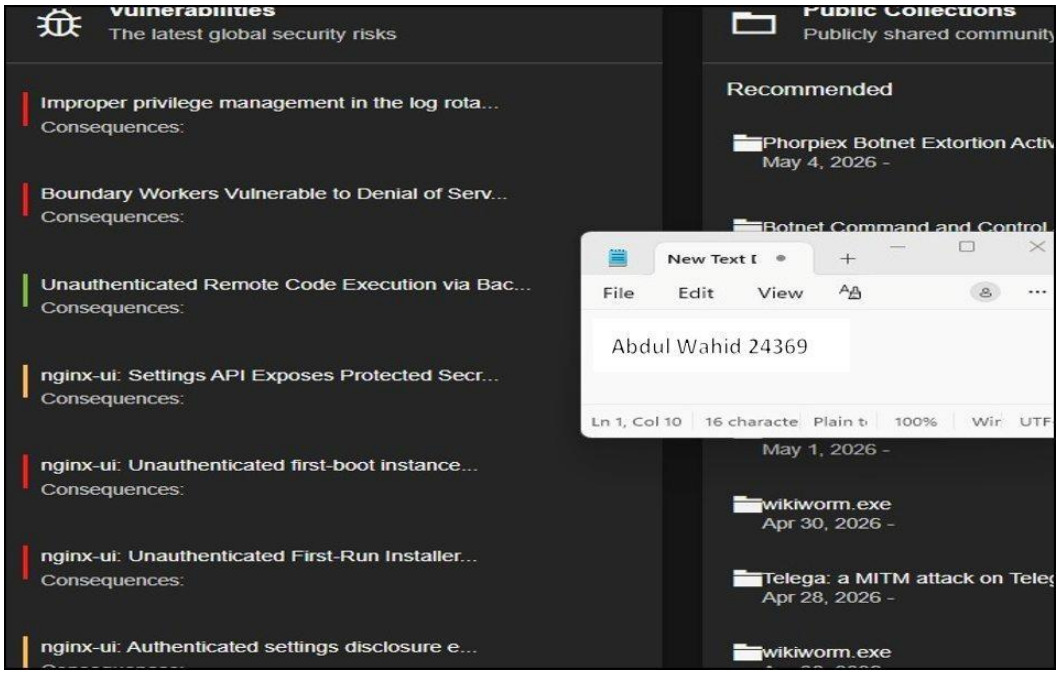


Fig 1.11 – Vulnerability Detail: CVE-2026-42221 (CVSS 8.1 High) – nginx-ui Unauthenticated Admin Claim

Step 8: Security Intelligence Blog

Visited the Security Intelligence Blog section from the dashboard to collect additional intelligence and insights on information security published by IBM analysts.

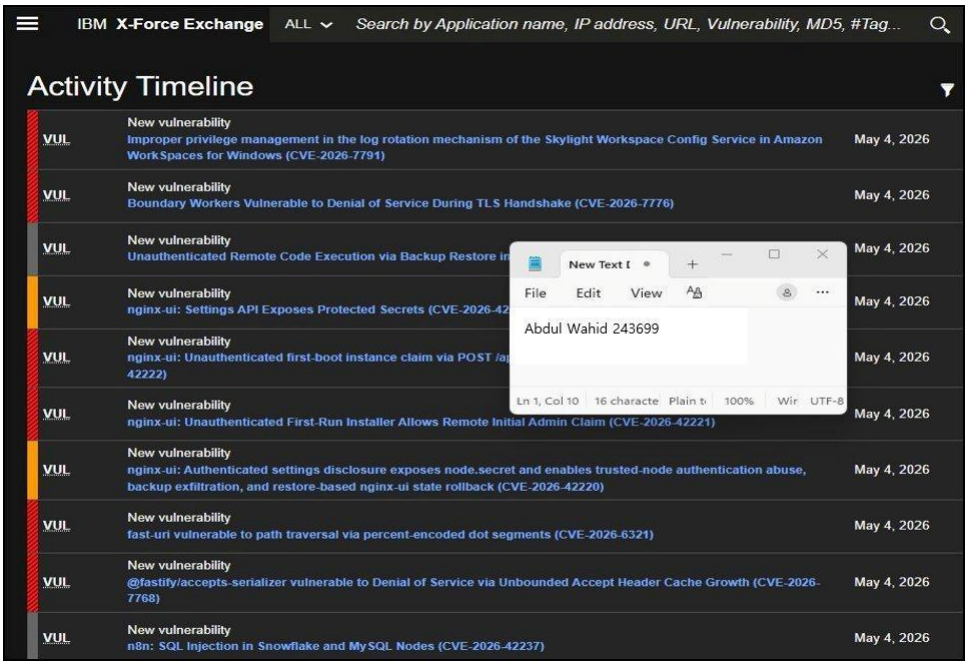


Fig 1.12 – Security Intelligence Blog Section

Step 9: Botnet Distribution – qsnatch_a

Navigated to the Botnet Distribution section. Viewed details for the trending botnet "qsnatch_a" including affected countries (121 countries), peak date (Apr 9, 2026), and global infection map.

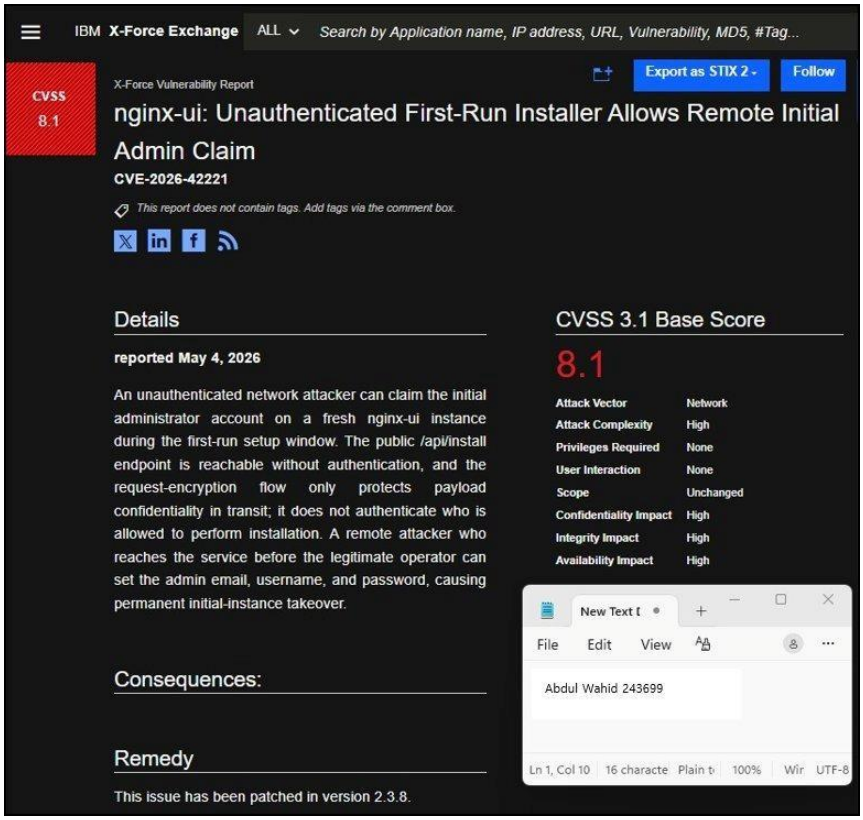


Fig 1.13 – Botnet Distribution: qsnatch_a (121 Affected Countries, Peak Apr 9, 2026)

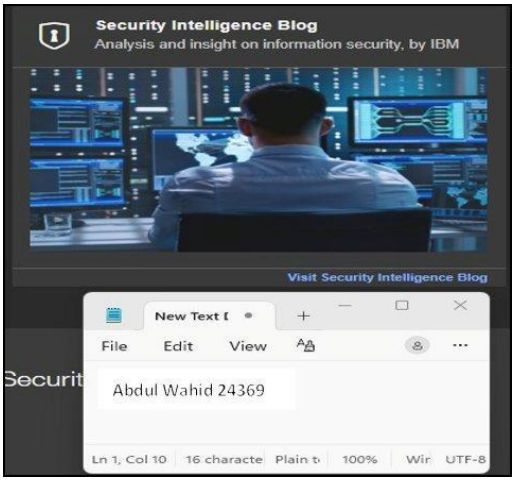


Fig 1.14 – Trending Botnets List with qsnatch_a Selected

Task 2: Threat Analysis using IBM X-Force Exchange

Step 1: Scanning a Malware File

Used the "Scan file" feature on the IBM X-Force Exchange Dashboard to upload and scan a malware sample. The scanned file was identified and analyzed automatically.

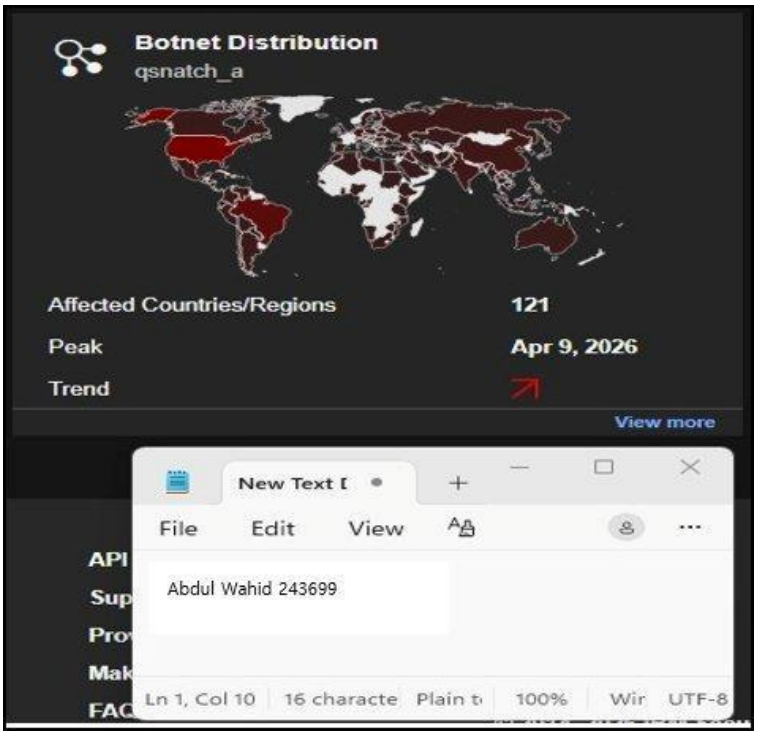


Fig 1.15 – Scan File Feature on IBM X-Force Exchange Dashboard

Step 2: Malware Analysis Results

The scanned file produced an X-Force Malware Report showing the MD5 hash. The analysis identified the malware as belonging to the "tedy" family, classified as a Trojan, with Risk Level: High. Community coverage was 64%, platform: Win64.

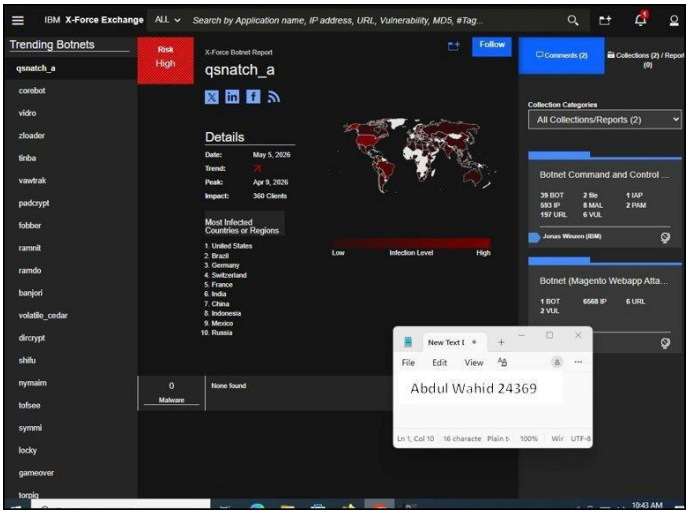


Fig 1.16 – X-Force Malware Report: Hash b7476812248be874309354a96b46438b, Family: tedy, Type: Trojan, Risk: High

Task 3: Threat Intelligence Sharing using IBM X-Force Exchange

Step 1: Creating a New Collection

Navigated to Collections via the menu and created a new collection named "torjan" to document and share the analyzed threat intelligence.

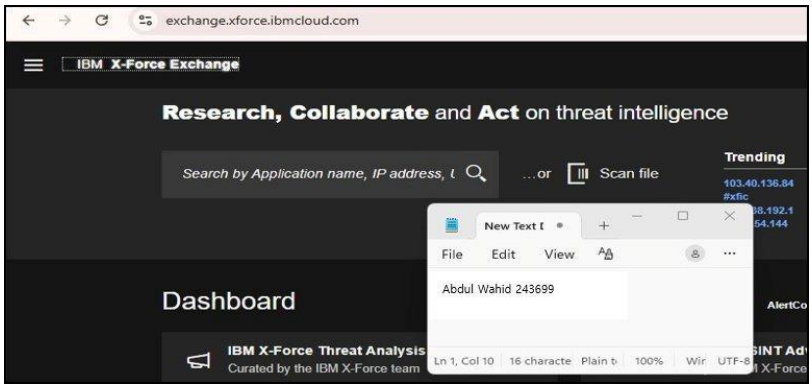


Fig 1.17 – Collections Page with New Collection "torjan" Created

Step 2: Adding Threat Intelligence Details

Edited the collection to fill in threat intelligence details including Threat Type, Overview, Indicators of Compromise, Protection measures, and Recommendations.

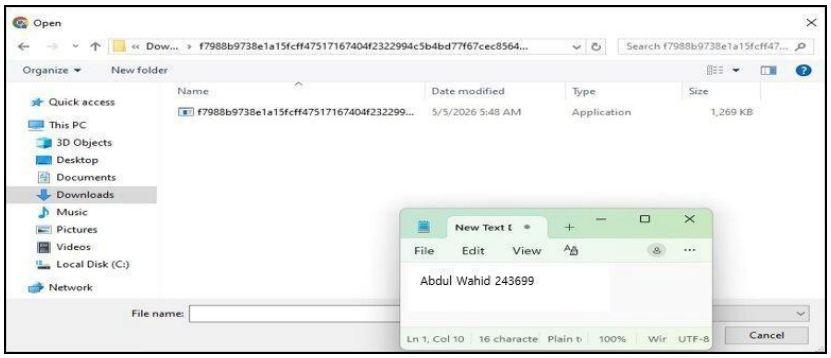


Fig 1.18 – Editing Collection: Adding Trojan Threat Details and IoCs

Step 3: Setting Sharing Options

Configured Sharing Options to share the collection publicly with others using TLP: WHITE protocol and set Public (anyone on the internet) access to "Contribute".

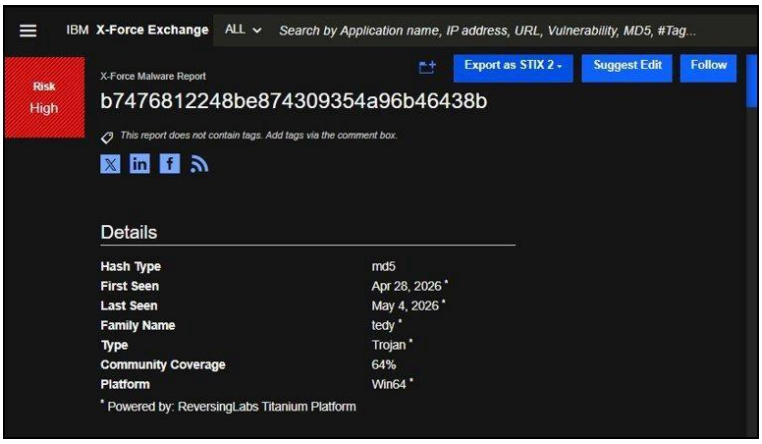


Fig 1.19 – Sharing Options: Share with Others, TLP WHITE, Public Contribute

Summary – Lab 5

Successfully completed all three tasks of Lab 5 using IBM X-Force Exchange:

- Task 1 – Data Collection: Collected IoC data (SHA-256 hashes, IPv4 addresses), vulnerability details (CVE-2026-42221, CVSS 8.1), botnet distribution data (qsnatch_a), and current malicious activity information.
- Task 2 – Threat Analysis: Analyzed a malware file using the Scan file feature; identified it as a Trojan (tedy family) with High risk and MD5 hash b7476812248be874309354a96b46438b.
- Task 3 – Threat Intelligence Sharing: Created and published a threat intelligence collection named "torjan" publicly with TLP: WHITE for community collaboration.

Lab 11: Threat Intelligence Reporting using KeepNote & Microsoft Word

This lab demonstrates threat intelligence report generation using KeepNote (a note-taking application) and threat intelligence report writing using Microsoft Word. The reports document findings from previous lab tasks.

Task 1: Threat Intelligence Report Generation using KeepNote

Step 1: Installing KeepNote

Downloaded and installed KeepNote from the provided installer. Followed the setup wizard with default options, installing to C:\Program Files (x86)\KeepNote.

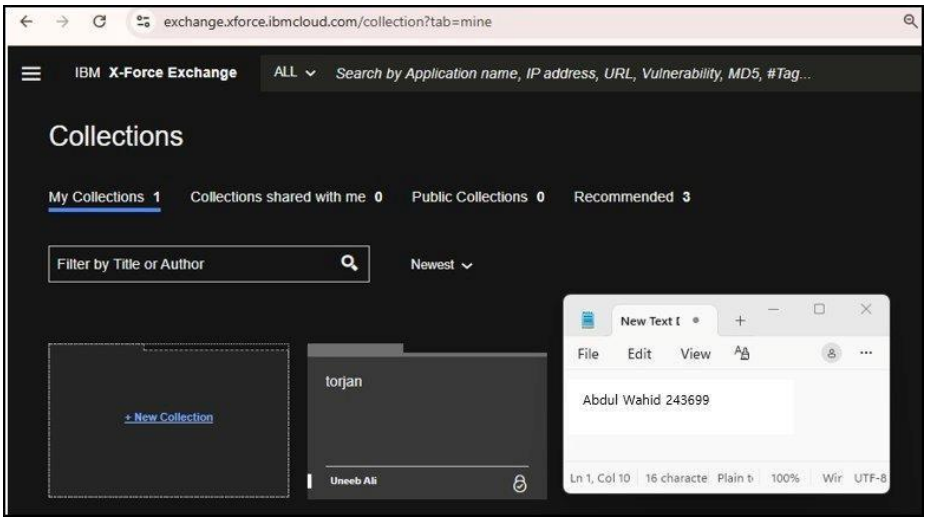


Fig 2.1 – KeepNote Setup Wizard – Select Destination Location

Step 2: Launching KeepNote

After installation, KeepNote launched automatically displaying the main console ready for notebook creation.

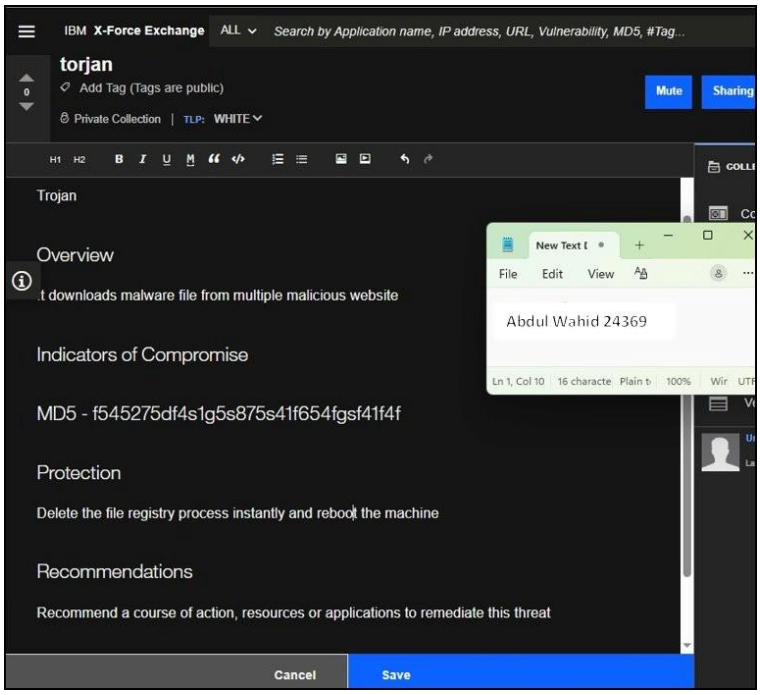


Fig 2.2 – KeepNote Application Main Console

Step 3: Creating a New Notebook

Created a new notebook named "sample Notebook" using File → New Notebook, saved in the default KeepNote folder.

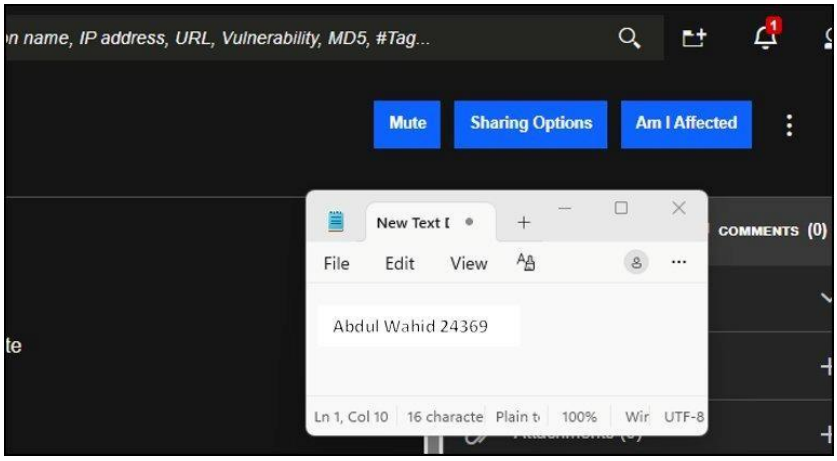


Fig 2.3 – Creating New Notebook: "sample Notebook"

Step 4: Adding Folder – Data Collection Through DNS

Right-clicked on the Sample Notebook root and added a new folder named "Data collection through DNS" to organize DNS interrogation findings.

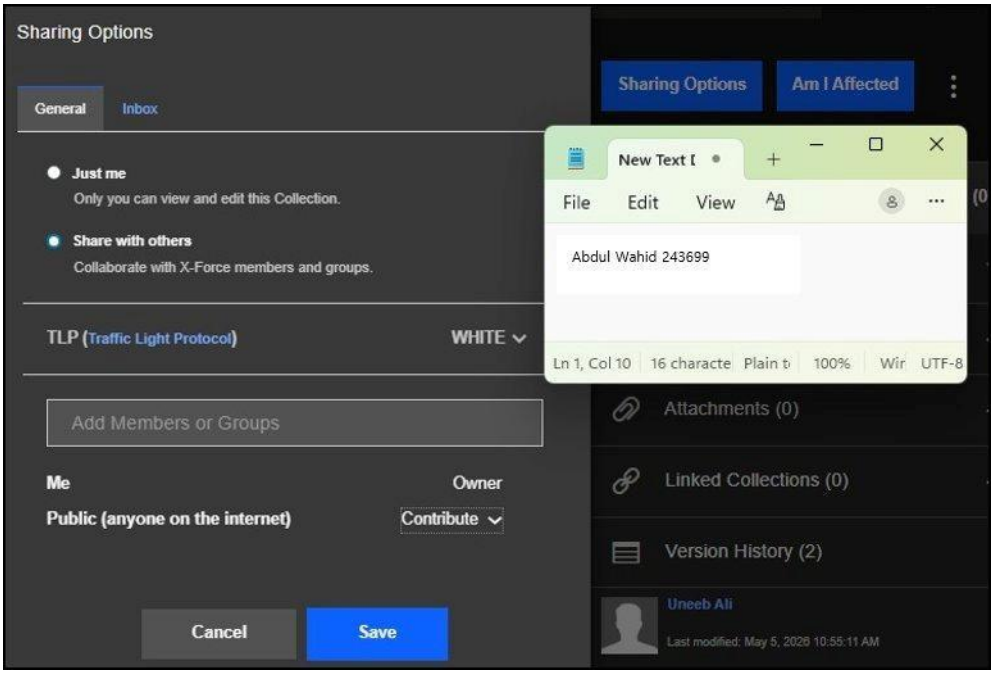


Fig 2.4 – Sample Notebook with "Data collection through DNS" Folder Created

Step 5: Adding Folder – Gathering Organization IP and Domain

Added another folder inside the Data Collection folder named "gathering organization ip and domain" as a subfolder to organize IP and domain name intelligence.

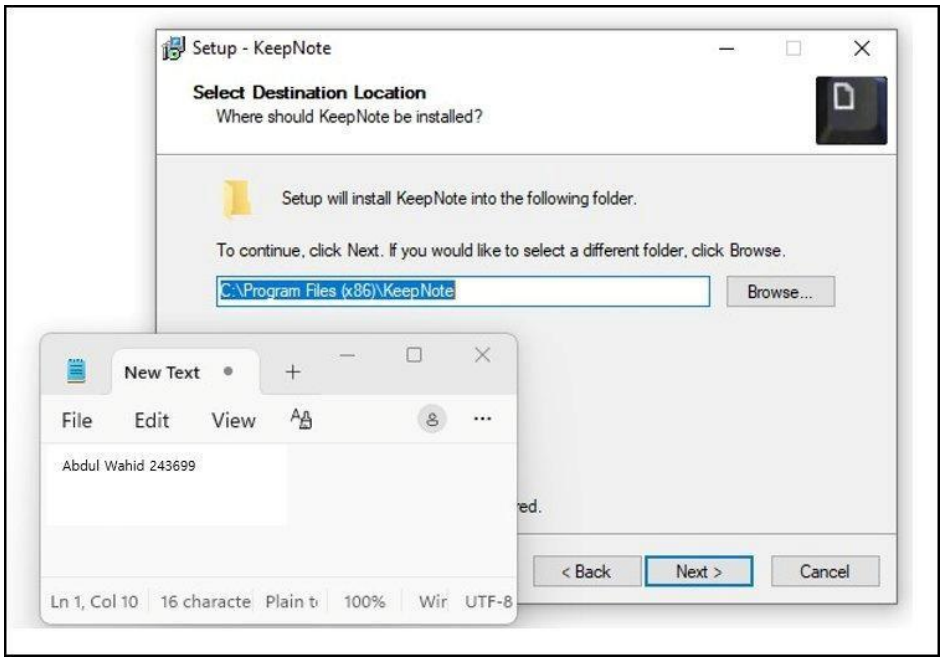


Fig 2.5 – "Gathering Organization IP and Domain" Sub-folder Added

Step 6: Adding Child Page – Whois Lookup

Right-clicked the "gathering organization ip and domain" folder and added a New Child Page named "whois lookup" to document the WhoIs lookup results.

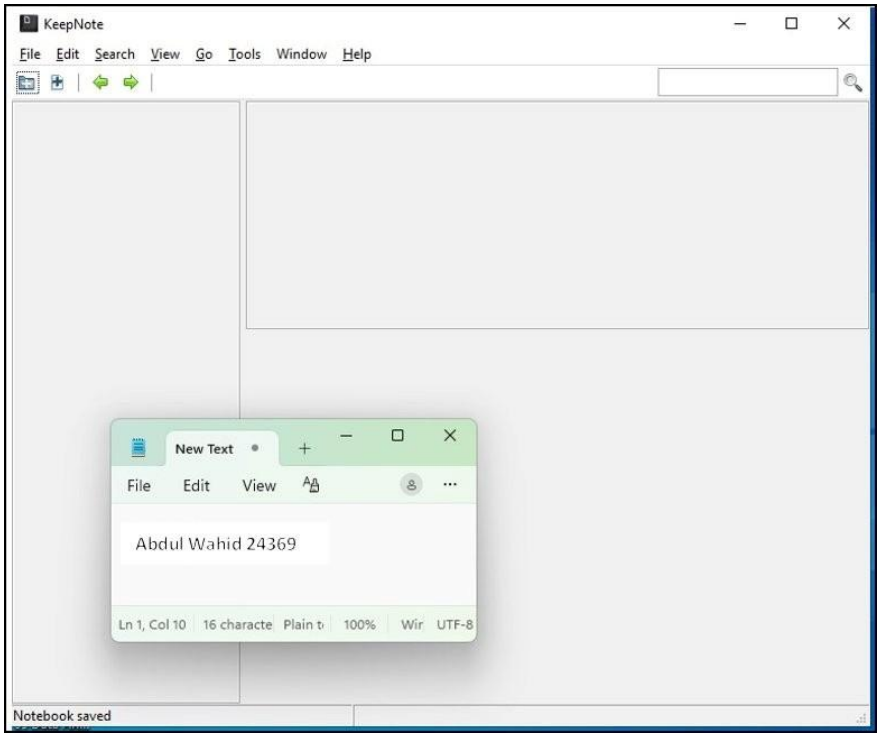


Fig 2.6 – WhoIs Lookup Child Page Added under Gathering Organization IP Folder

Step 7: Documenting WhoIs Lookup Results

Filled in the WhoIs Lookup page with the task details and results obtained from the WhoIs lookup on www.certifiedhacker.com, including name servers, IP address, IP location, and domain status.

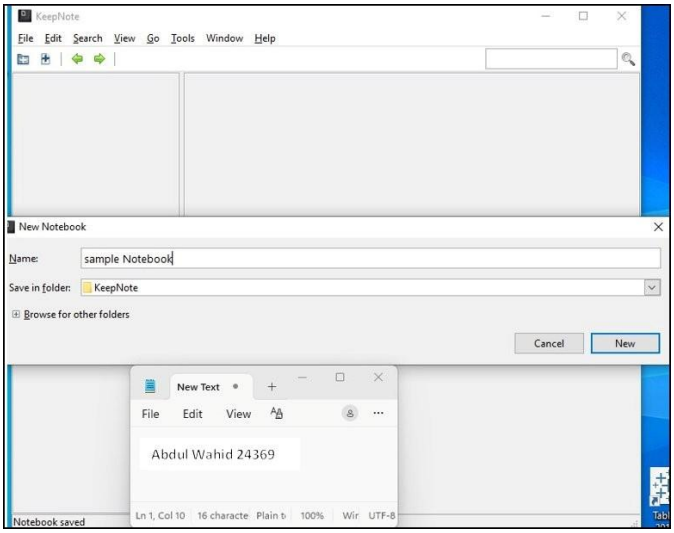


Fig 2.7 – WhoIs Lookup Page with Complete Data Entry

Step 8: Exporting Notebook as HTML

Exported the complete notebook to HTML format using File → Export Notebook → HTML. Saved the exported notebook to the Desktop as "sample Notebook-2026-05-05".

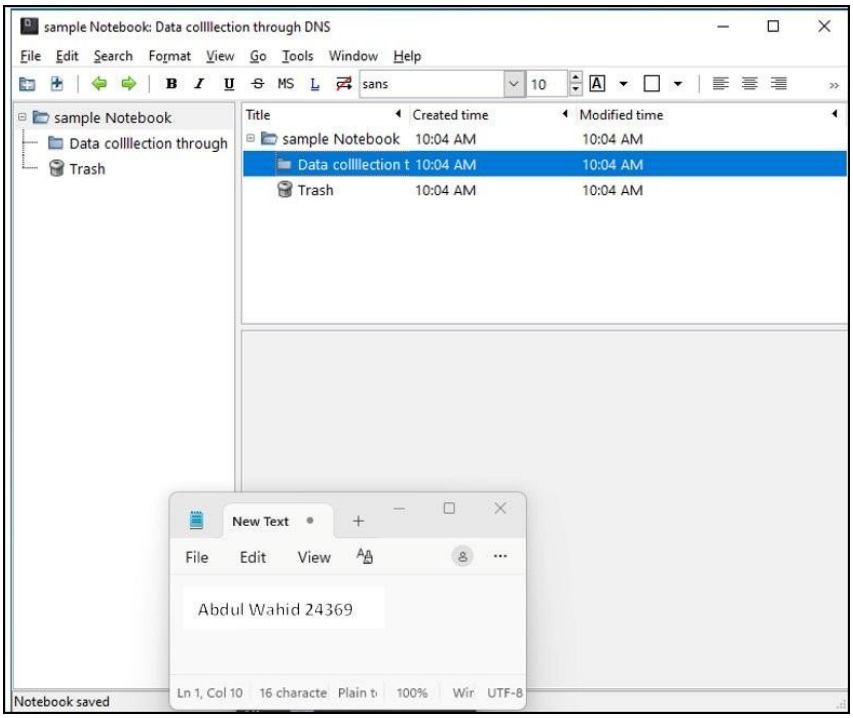


Fig 2.8 – Export Notebook Dialog – Saving to Desktop

Step 9: Viewing Exported Files

Navigated to the exported folder on the Desktop and confirmed the presence of all exported files including index.html, node, notebook, tree, and the data folders.

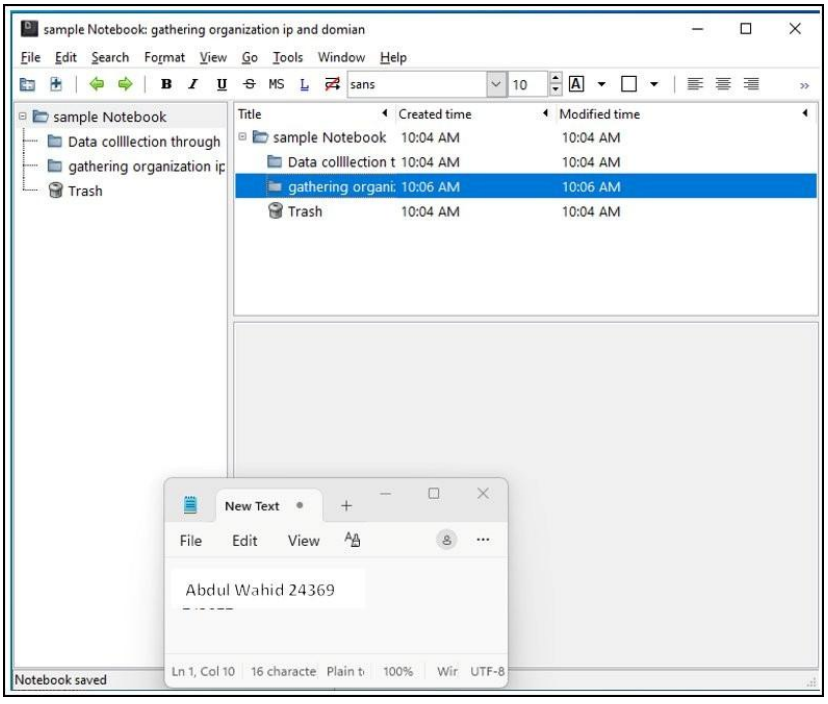


Fig 2.9 – Exported Notebook Folder Contents on Desktop

Step 10: Opening Report in Browser

Double-clicked the index.html file to open the exported report in a web browser. The report displays the notebook structure in the left pane and content in the right pane, including all WhoIs lookup data.

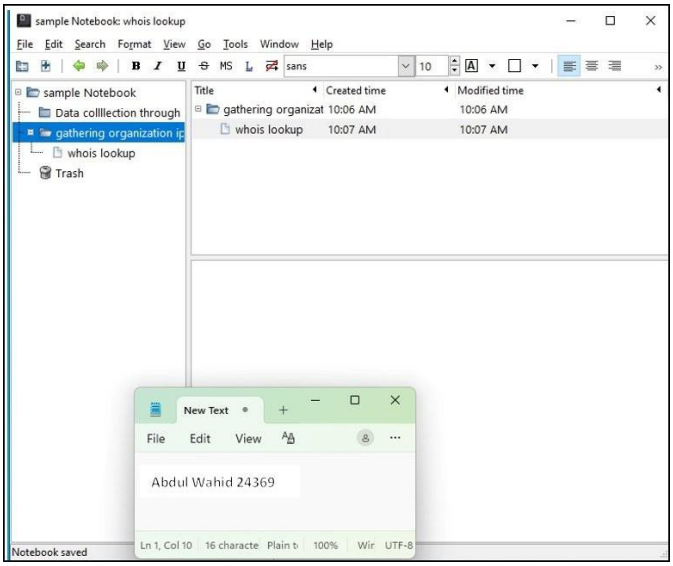


Fig 2.10 – KeepNote Report Viewed in Browser (WhoIs Lookup Data Visible)

Step 11: Viewing the Full CTIA Lab Report

Opened the CTIA Lab Report index.html to view the complete threat intelligence report for all lab modules, including Data Collection, Data Analysis, and Intelligence Reporting sections.

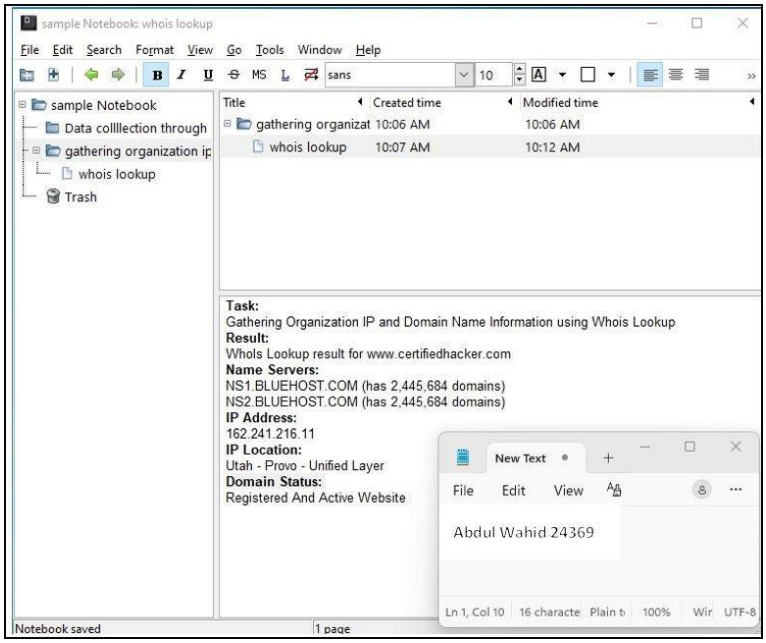


Fig 2.11 – Full CTIA Lab Report Viewed in Browser with Complete Tree Structure

Task 2: Threat Intelligence Report Writing using Microsoft Word

Step 1: Opening the Cyber Threat Intelligence Report Template

Navigated to the TI Report folder and opened the Cyber Threat Intelligence Report Template.docx file in Microsoft Word to begin filling in the threat intelligence report.

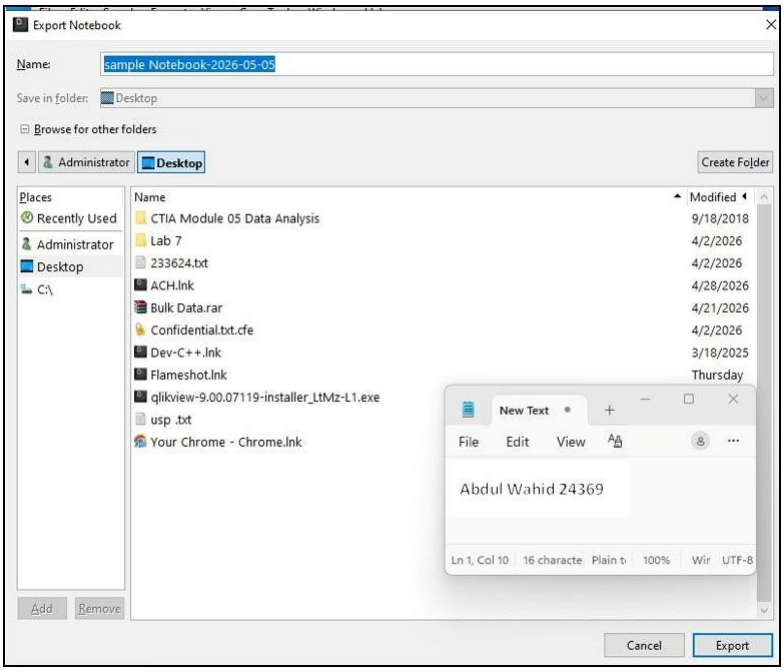


Fig 2.12 – TI Report Folder with Template and Sample Report Files

Step 2: Reviewing the Completed Cyber Threat Intelligence Report

Opened the completed Cyber Threat Intelligence Report.docx which serves as a reference. The report includes Report Details (Author, Company, Date), Client Details (Name, Company, Industry, Location), and Test Details sections fully filled in.

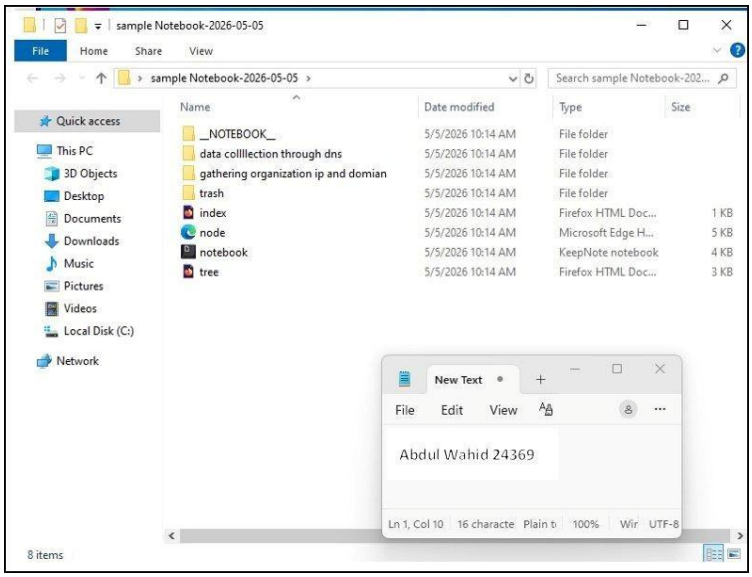


Fig 2.13 – Cyber Threat Intelligence Report.docx – Report Details and Client Details Tables

Summary – Lab 11

Successfully completed both tasks of Lab 11:

- Task 1 – KeepNote Report Generation: Installed KeepNote, created a structured notebook with folders for Data Collection Through DNS and Gathering Organization IP and Domain, documented WhoIs lookup findings for www.certifiedhacker.com, and exported the report in HTML format viewable in a web browser.
- Task 2 – Microsoft Word Report Writing: Opened and reviewed the Cyber Threat Intelligence Report Template, examined the completed sample report which documents findings from all modules including report details, client information, and comprehensive test analysis.

Lab Analysis & Conclusion

This lab exercise covered two key aspects of the Threat Intelligence lifecycle:

IBM X-Force Exchange (Lab 5)

IBM X-Force Exchange proved to be a powerful all-in-one threat intelligence platform. Through its use, the following data was successfully collected and analyzed:

- Indicators of Compromise (SHA-256 hashes and IPv4 addresses) from the "New Tactics Seen In Smoke Loader" advisory with 28 associated reports.
- Current malicious activity across the globe via the interactive threat activity map.
- Vulnerability intelligence including CVE-2026-42221 (CVSS 8.1 High) related to nginx-ui unauthenticated admin claim.
- Botnet distribution data for qsnatch_a affecting 121 countries with peak on Apr 9, 2026.
- Malware analysis results classifying a scanned file as a Trojan (tedy family, Risk: High) using the MD5 hash b7476812248be874309354a96b46438b.
- Successfully created and publicly shared a threat intelligence collection ("torjan") with TLP: WHITE, enabling community collaboration.

Threat Intelligence Reporting (Lab 11)

KeepNote provided an effective structured reporting tool for organizing and documenting intelligence findings:

- Created a hierarchical notebook structure mirroring the threat intelligence collection methodology.
- Documented WhoIs lookup results for www.certifiedhacker.com: IP 162.241.216.11, hosted on NS1/NS2.BLUEHOST.COM, located in Utah - Provo - Unified Layer, Domain Status: Registered And Active Website.
- Exported the notebook to HTML format for easy sharing and web-based viewing.
- The Microsoft Word Cyber Threat Intelligence Report template provides a professional standardized format for formal intelligence dissemination to stakeholders.

Overall, these labs demonstrate the complete threat intelligence lifecycle from data collection and analysis through to structured reporting and dissemination – core competencies for a Certified Threat Intelligence Analyst.